

Software Requirements Specification (SRS) – SKILL MENTOR:

DEFINITION:

A software requirements specification (SRS) is a document that describes what the software will do and how it will be expected to perform. It also describes the functionality the product needs to fulfill all stakeholders (business, users) needs.

PURPOSE:

The purpose of this SRS document is to provide a detailed overview of our software product, its parameters and goals. This document describes the project's target audience and its user interface, hardware and software requirements. It lays out the functional and non-functional requirements of a system that is used for describing the user interactions.

Template of SRS:

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SRS For SKILL MENTOR:

1. Introduction

Skill Mentor is a personalized skill learning and mentoring platform designed for students and self-learners. Unlike traditional e-learning platforms that focus purely on content delivery, Skill Mentor acts as a personal AI mentor for every learner, guiding them through learning, practice, revision, and assessment for each skill they choose to develop. Through Skill Mentor, learners interact with a dedicated AI assistant that provides curriculum-aligned explanations, adaptive assignments, and revision aids for every subject or skill they are studying.

2. Purpose

This document is intended for the following group of people: -

- Developers for the purpose of design, development, maintenance, and future enhancement of the software.
- Documentation writers.
- Project mentors, evaluators, and academic reviewers.
- Testers.

3. Scope

This document applies to the Skill Mentor AI software. This software offers benefits such as AI-guided skill learning, instant doubt clarification, adaptive assignment generation, automatic mind map and mnemonic creation, and collaborative learning with friends. It also allows learners to combine two or more related skills to receive interdisciplinary learning recommendations.

The software takes as input the learner's registered account credentials and their skill selection for login. The outputs then comprise an interactive dashboard that lets the learner select the skill, ask doubts, receive assignments, and access learning aids for that session.

The software is expected to be completed within a duration of 6 months as an academic project, developed and maintained by the project team.

4. Definitions, Acronyms, and Abbreviations

AI Artificial Intelligence

API Application Programming Interface

LMS Learning Management System used to organise and deliver structured skill content

NLP Natural Language Processing, used to interpret and respond to learner questions CMS Content Management System used to store and manage textbooks and reference material

ML Machine Learning; UI A visual interface through which the learner interacts with the platform; Session Log A time-stamped record of a learner's activity, used for progress tracking.

MB Mega Bytes

Ms Milliseconds.

Sec Seconds

5. References

The references for the above software are as follows: -

1. www.google.co.in
2. www.wikipedia.com

6. Overview

Section 1.0 discusses the purpose and scope of the software.

Section 2.0 describes the overall functionalities and constraints of the software and user characteristics. Section 3.0 and 4.0 details all the requirements needed to design the software.

7. The Overall Description

8. Product Perspective

Skill Mentor is a single, unified learning ecosystem consisting of various functional subcomponents. This software allows the learner to access a personalised AI mentor for every skill remotely through a web application. This software also allows the learner to perform various other functions apart from just learning a skill, such as doubt clarification, assignment generation, and revision aid creation. Some of its core software components are the skill-tracking module, the AI mentor engine, the assignment generator, the learning-aid generator, the collaboration module, and the analytics dashboard.

9. Product Functions

The major functions that Skill Mentor performs are described as follows: -

- **Skill Selection:** The dashboard provides the learner with a list of skills from which they can select any one in order to begin a learning session. After skill selection, the learner is prompted with an option to set the selected skill as their default focus so that they are not shown the selection screen in future sessions, thus making the experience a bit faster. The learner also has the freedom to switch to a different skill mentioned in the list at any point during that session.
- **Skill Profile Management:** The various functions that a learner can perform within their skill profile are as follows: - **Proficiency Level:** - The learner has the freedom to set or update their current proficiency level for the selected skill, i.e., whether they are a beginner, intermediate, or advanced learner.
- **Doubt Clarification:** The software allows the learner to type or select the doubt they want clarified, and the AI mentor responds with a step-by-step, example-based explanation aligned with the reference textbook for that skill.
- **Assignment Generation:** The proficiency level and topic for which the learner wants an assignment is then specified by the learner, and the system generates practice questions that progress from easy to advanced.
- **Learning Aid Generation:** Learning aid generation shall be done on request. After selecting the topic and the type of aid required, the learner receives a mind map, mnemonic, or short summary that is generated and displayed within the session.
- **Multi-Skill Linking:** Multi-skill linking shall be facilitated between any two or more skills added to the learner's profile, for example linking a programming skill and a mathematics skill for interdisciplinary assignments.

- Progress Enquiry: Progress enquiry for any skill linked to the learner's profile shall be facilitated.
- Reporting: Every completed assignment or activity shall be recorded in the learner's history and reflected on their dashboard. The reporting module enables the learner to choose whether they want a detailed activity summary or just an update to their overall skill progress.
- Session Cancelling: The learner shall abort an ongoing assignment or activity with the press of an Exit option. For example, on starting an assignment by mistake. In addition, the learner can also cancel the entire session by logging out and can start a fresh session all over again.

10. User Characteristics

There are different kinds of users that will be interacting with the system. The intended users of the software are as follows: -

- User A: A beginner learner. This user has little or no prior exposure to the selected skill and is not a frequent user of AI-based learning tools. User A will find the product easy to use due to simple, guided explanations and beginner-level assignments for each skill.
- User B: An experienced learner. This user has used the platform on several occasions before and manages most of their skill development independently through Skill Mentor. There is only a brief onboarding session at the beginning, thus making subsequent sessions faster.
- Platform Administrator: A system maintainer. This user is familiar with the functioning of the platform. This user is in charge of updating the reference material and reference textbooks used by the AI mentors and resolving technical issues in case of malfunction. This user is presented with a different dashboard when they log in with the administrator's password and is provided with options different from those of a normal learner. They have the authority to add, update, or restrict skills and content available on the platform.

11. Constraints

The major constraints that the project has are as follows: -

- The platform must support only one active learning session per learner account at a time.
- The number of invalid login attempts must not exceed five. After five unsuccessful attempts, the account is temporarily locked and needs to be unlocked through email verification.
- The minimum number of questions generated per assignment request is 5, and the maximum number of questions generated in a single request is 20, with a maximum of 100 questions that a learner can attempt across all skills in a single day.
- Before a new assignment is generated, a check is performed to confirm that the learner has completed at least 70% of the previous assignment for that skill, failing which the new assignment request is denied.
- The minimum length of a doubt query submitted by the learner is 5 characters, and the maximum length is 500 characters. A learner can link only those external practice platforms that are supported by the system, such as LeetCode, HackerRank, and GeeksforGeeks.

12. Assumptions and Dependencies

The requirements stated in the SRS could be affected by the following factors:

- One major dependency that the project might face is the changes that need to be incorporated as academic curricula and reference textbooks are revised. As the source material changes, the AI mentor's knowledge base needs to be updated with the same immediately. A delay in doing so may result in explanations that no longer match the prescribed material. So, the reference content should be refreshed as and when required by the content team.
- Another constraint relating to the operating environment is that we are specific to a cloud-hosted PostgreSQL database.
- The project could be largely affected if a learner's progress is updated from one device at the same time the learner is active on another device. Such a condition shall be taken care of through real-time data synchronization.
- At this stage no quantitative measures are imposed on the software in terms of speed and memory although it is implied that all functions will be optimized with respect to speed and memory.

It is furthermore assumed that the scope of the platform will increase considerably in the future.

13. System Features

3.1 Functional requirements

Personalized Skill Learning and Mentoring

At the start, the learner is provided with a login screen and is required to enter their registered email and password, which will be verified by the system. In case of an unsuccessful attempt, the learner is asked again for their credentials, but the maximum number of attempts given to the learner is limited to 5 only, failing which the account is temporarily locked and needs to be unlocked through email verification.

After logging in, the learner is presented with a list of skills added to their profile. The learner can select any one skill for interaction with its dedicated AI mentor for that session. 1. There is also a facility for the learner to switch to any other skill during that session. 6

After skill selection, the learner is directed towards a skill dashboard that displays a set of options, 1. such as Ask a Doubt, 3. Get an Assignment, and 5. Generate Learning Aid, along with a brief description of each, enabling the learner to understand their functioning. The learner can select any of the listed options and can continue with the learning activity.

At any moment if the learner wants to abort the current activity, they are provided with an Exit option. Just by selecting the exit option, 9 the learner can cancel the changes made so far in that activity and can begin a new one. After the learner is finished with their session, they are required to log out of the platform. 9

Validity Checks

In order to gain access to the system, the learner is required to enter their correct registered email and password, failing which their account may be temporarily locked. The learner can access only one account at a time. Also, if the learner is an administrator, they are required to enter their administrator credentials in order to access and change the content and settings provided by the system.

Sequencing Information

The information about the learner and their skill profile should be entered into the database prior to any learning activity, 7. and a backup should be maintained for all learner progress information.

Error Handling/ Response to Abnormal Situations

If any of the above validation/sequencing flow does not hold true, appropriate error messages will be prompted to the learner for taking the needful corrective action.

4. External Interface Requirements

1. User Interface Requirements

1. A login screen is provided in the beginning for entering the required registered email and password.
2. An unsuccessful login leads to a reattempt (maximum five) screen for again entering the same information. A successful login leads to a screen displaying the list of skills added to the learner's profile from which the learner can select any one.
3. In case of an administrator, a screen will be shown having options to add a new skill, update reference content, manage learner accounts, and disable any service.
4. In case of content update or system maintenance, a screen is displayed to confirm the administrator's intent to proceed and also allow the administrator to take a backup of the data if needed.
5. After skill selection, a screen with a number of options is then shown to the learner. It contains all the options, such as doubt clarification, assignments, and learning aids, along with their brief description to enable the learner to understand their functioning and select the proper option.
6. A screen will be provided for the learner to check their overall skill progress.
7. A screen will be provided that displays the list of friends the learner is connected with and their shared learning progress.
8. A screen will be provided for the learner to link two or more skills and view combined, interdisciplinary assignment recommendations.

Other various user interface requirements that need to be fulfilled are as follows:-

The display shall be responsive and shall render correctly on screen resolutions of 1366 x 768 and above, as well as on mobile screens of 360 pixels width and above. The display shall support a light and dark colour theme. The interface shall also support touch-screen interaction on tablets and mobile browsers. The dashboard shall use accessible font sizes and colour contrast suitable for extended reading sessions.

2. Hardware Interface Requirements

- The client device shall have a minimum screen size of 5 inches to properly access the responsive interface. The device shall support a modern web browser released within the last three years. A stable internet connection with a minimum bandwidth of 1 Mbps is recommended for uninterrupted use of the AI mentor and doubt-clarification features.

• A standard keyboard, mouse, or touch-screen input device is required

• A working microphone is optional and required only for learners who c
input.

• There shall be sufficient local storage available on the client device f
notes and mind maps.

• The server infrastructure shall provide a minimum of 16 GB of RAM and
accommodate an increasing number of concurrent learners. Screen resolution of at least 1024 x 768 is
required on the administrator console for proper and complete viewing of management screens.

3. Software Interface Requirements

In order to perform various different functions, this software needs to interact with various other softwares. So there are certain software interface requirements that need to be fulfilled which are listed as follows:-

The backend service used to manage learner data, skill tracking, and assignment logic shall be a Node.js-based REST API. The AI mentor engine used to generate explanations, assignments, and learning aids shall be built on a large language model accessed through a secured API. The database used to keep records of learner accounts and progress shall be PostgreSQL version 14 or higher. The frontend interface shall be built using a modern JavaScript framework such as React.

4. Communication Interface Requirements

The machine needs to communicate with the main branch for each session for various functions such as login verification, account access etc. so the following are the various communication interface requirements that are needed to be fulfilled in order to run the software successfully: -

The system will employ HTTPS for all client-server communication to ensure secure data transfer. The communication protocol used shall be TCP/IP. The AI mentor and doubt-clarification features shall use a REST API for standard requests and a WebSocket connection for real-time chat-style interactions.

1. Other Non-functional Requirements

1. Performance Requirements

The following list provides a brief summary of the performance requirements for the software:

2. Capacity

The platform shall provide learners with 24-hour access to skill content, assignments, and the AI mentor.

3. Dynamic requirements

The AI mentor's response time for a doubt-clarification query must not exceed 3 sec. under normal server workload and 5 sec. under peak server workload. The assignment generation time must not exceed 4 sec. under normal server workload and 6 sec. under peak server workload. Learning-aid generation time (mind maps, mnemonics, summaries) must not exceed 5 sec. under normal server workload and 8 sec. under peak server workload. Dashboard and progress-analytics load time must not exceed 3 sec. under normal server workload and 5 sec. under peak server workload. Login and skill-selection screen transitions must not exceed 2 sec. under normal server and peak server workload.

4. Quality –

The primary objective is to produce quality software. As the quality of a piece of software is difficult to measure quantitatively, the following guidelines will be used when judging the quality of the software:

1. Consistency – All code will be consistent with respect to the style. (This is implied when adhering to the standard).
2. Test cases – All functionality will be thoroughly tested

5. Software System Attributes

1. Reliability

- The AI mentor's explanations shall be generated from verified textbooks and reference material to ensure accuracy and reduce the chances of incorrect or misleading responses. The learner progress data shall be stored using a reliable, replicated database system.

2. Availability

The platform shall have a backup data store in case of primary database failure. Any abnormal operation shall result in a graceful error message rather than a silent failure.

After an abnormal shutdown of the AI mentor service, the system shall automatically attempt to restart the affected service, with manual intervention available for the maintenance team if needed.

There should be no loss of learner progress data during whose update the system is abnormally interrupted.

3. Security

The system shall be compatible with standard web-application security practices. The system shall have two levels of security, i.e., account login and session-token verification, both authenticated by the backend service. The encryption standard used during data transmission shall be TLS 1.2 or higher.

The password shall be 8-20 characters long. Passwords shall not contain the learner's name, as such passwords are easy to guess. Passwords can contain digits, hyphens, and underscores.

The learner should be provided with only five attempts for login, failing which the account needs to be temporarily locked and unlocked through email verification.

There shall be activity logging for all administrator actions. There shall be a secured, access-controlled database for storing learner data. The learner data shall be handled in compliance with standard data-privacy practices.

4. Maintainability

The system components, i.e., the AI mentor engine, database, and API services, shall be easily serviceable without requiring downtime for the entire platform.

The system should have the mechanism of self-monitoring periodically in order to detect any fault in the AI mentor engine or database connectivity.

The system should inform the maintenance team automatically as soon as it detects any error. The kind of fault and the problem being encountered should also be reported by the system automatically.

6. Other Non-functional Requirements

6.1 Safety Requirements

The following safety requirements shall be fulfilled by the Skill Mentor platform:

- The system shall automatically save learner progress at regular intervals to prevent data loss during unexpected failures.
- The learner shall receive a confirmation message before exiting an ongoing assignment or deleting any saved information.
- The system shall recover previously saved learner data after an abnormal shutdown.
- Periodic backups of learner records, assignments, and progress data shall be maintained.
- Only authorized administrators shall be allowed to modify reference content or system configurations.
- The platform shall generate appropriate warning messages whenever critical operations are performed.
- In case of network interruption, the learner shall be able to continue from the last successfully saved state after reconnection.

Software Quality Attributes

Usability

- The platform shall provide an intuitive and user-friendly interface.
- The navigation shall remain consistent throughout the application.
- First-time users shall be able to use the platform with minimal guidance.
- The system shall support both light mode and dark mode.
- The interface shall be accessible on desktop, tablet, and mobile devices.

Scalability

- The system shall support an increasing number of learners without significant degradation in performance.
- Additional skills and AI mentor modules shall be integrated without modifying the existing architecture.
- The backend shall support horizontal scaling for handling concurrent users.

Portability

- The application shall operate on Windows, Linux, and macOS through modern web browsers.
- The platform shall support Android and iOS mobile browsers.
- The software shall remain independent of platform-specific hardware.

Interoperability

- The system shall support integration with external learning platforms such as LeetCode, HackerRank, and GeeksforGeeks.
- The platform shall communicate with third-party services through secure REST APIs.
- Data exchange shall use standard JSON format.

Business Rule

The Skill Mentor platform shall comply with the following business rules:

- Every learner shall register using a unique email address.
- Only registered learners shall access learning services.
- Only one active session shall be allowed for each learner account.
- Assignment difficulty shall be determined according to the learner's selected proficiency level.
- A learner shall complete at least 70% of the previous assignment before requesting a new assignment for the same skill.
- Only administrators shall be authorized to add, modify, or remove learning content.
- All learner activities shall be automatically recorded in the progress history.
- Every administrator action shall be logged for auditing purposes.

Other Requirements

The following additional requirements shall also be satisfied:

- The platform shall provide email notifications for account verification, password reset, and important announcements.
- Learners shall be able to export their progress reports in PDF format.